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Original Article

Virus-free CRISPR knockin of a chimeric antigen receptor into *KLRC1* generates potent GD2-specific natural killer cells

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Natural killer (NK) cells are an appealing off-the-shelf, allogeneic cellular therapy due to their cytotoxic profile. However, their activity against solid tumors remains suboptimal in part due to the upregulation of NK-inhibitory ligands, such as HLA-E, within the tumor microenvironment. Here, we utilize CRISPR-Cas9 to disrupt the *KLRC1* gene (encoding the HLA-E-binding NKG2A receptor) and perform non-viral insertion of a GD2-targeting chimeric antigen receptor (CAR) within NK cells isolated from human peripheral blood. Genome editing with CRISPR-Cas9 ribonucleoprotein complexes yields efficient genomic disruption of the *KLRC1* gene with 98% knockout efficiency and specific knockin of the GD2 CAR transgene as high as 23%, with minimal off-target activity as shown by CHANGE-seq, in-out PCR, amplicon sequencing, and long-read whole-genome sequencing. *KLRC1*-GD2 CAR NK cells display high viability and proliferation, as well as precise cellular targeting and potency against GD2$^+$ human tumor cells. Notably, *KLRC1*-GD2 CAR NK cells overcome HLA-E-based inhibition *in vitro* against HLA-E-expressing, GD2$^+$ melanoma cells. Using a single-step, virus-free genome editing workflow, this study demonstrates the feasibility of precisely disrupting inhibitory signaling within NK cells via CRISPR-Cas9 while expressing a CAR to generate potent allogeneic cell therapies against HLA-E$^+$ solid tumors.

INTRODUCTION

Chimeric antigen receptor (CAR) T cells, which involve genetic engineering by lentiviral or retroviral vectors of autologous peripheral blood mononuclear cells with synthetic receptors to CD19 or BCMA, have revolutionized treatment of blood cancers.[1] Despite high response rates ranging from 50% to 90%, CAR T cells have been shown to induce life-threatening toxicities like cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS) upon infusion.[2–5] As an alternative approach, CAR natural killer (NK) cells have displayed outstanding results in the clinic for treating lymphoid malignancies, with 30% of patients achieving a complete response with no instances of CRS or ICANS.[6,7] CAR NK cells have failed to produce comparable results in solid tumors in part due to the upregulation of NK-inhibitory checkpoints by the tumor.[8] One of these ligands, the nonclassical human leukocyte antigen (HLA) class I molecule HLA-E, is upregulated across several tumors and ligates the inhibitory NKG2A receptor as a heterodimer with CD94, limiting NK effector function.[9,10] NKG2A effectively serves as an NK immune checkpoint molecule that abrogates effector functions, such as cytotoxicity, in part by disrupting actin-NKG2D synapses.[11] Blockade of NKG2A by monoclonal antibodies, or ablation of the receptor through genomic engineering, has been shown to improve NK effector function *in vivo* as well as in clinical trials.[10,12–14] Combinatorial therapies utilizing blockade of NKG2A with CAR modification present a large potential for enhancing NK cell responses against solid tumors but have been unexplored to date.

As a component of the innate immune system, NK cells have high sensitivity to exogenous DNA—making genomic engineering of NK cells with viral vectors challenging, often resulting in low transduction efficiencies (<1%–20%) and poor cell viability.[15,16] In addition, viral vectors integrate randomly, leading to variability of transgene expression, with some risk of insertional mutagenesis.[17,18] Electroporation-based methods can deliver mRNA payloads in a non-viral manner quite efficiently, but only result in transient gene expression of the

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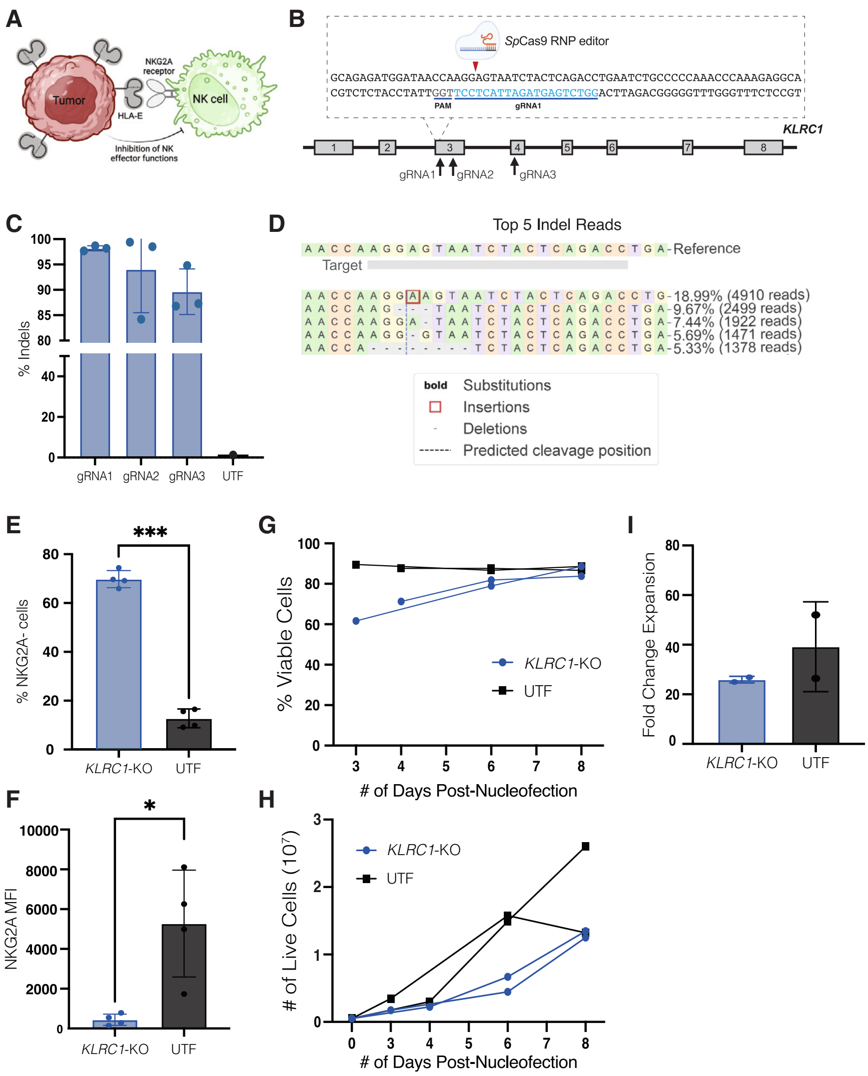


Figure 1. Cas9 ribonucleoprotein-based strategy yields high *KLRC1* knockout efficiency

(A) Schematic showing the interaction between NK cells and the tumor. The HLA-E ligand is often upregulated on the surface of tumors and inhibits NK effector functions by binding the inhibitory NKG2A receptor. (B) Schematic depicting knockout strategy. sgRNA1 (in blue), in complex with the *Sp*Cas9 editor, targets the third exon of the *KLRC1* gene. The locations of the three gRNA sequences are indicated by black arrows in sequential order. (C) Percent of indel formations for each of the three sgRNAs in NK-92 cells at the *KLRC1* gene. Data are shown for three replicates for one donor (n = 1). (D) Quantification of the top five allelic editing outcomes of gRNA1 as analyzed by CRISPResso2. (E) Primary PB-NK cells were nucleofected with RNP complexes targeting *KLRC1*. Percent of NKG2A$^-$ cells is shown in comparison to the UTF population, as measured by flow cytometry. Data are shown as the average of two replicates across four donors (n = 4). (F) MFI of the NKG2A receptor on the cell surface of *KLRC1*-KO and UTF cells. Data are shown as the average of two replicates across four donors (n = 4). (G and H) Cell viability and proliferation of *KLRC1*-KO and UTF cells after nucleofection. Data are shown as the average of two replicates across two donors (n = 2). (I) Calculated fold expansion of *KLRC1*-KO and UTF cells 8 days after nucleofection. A two-tailed paired t test and one-way ANOVA followed by Tukey's multiple comparisons test were used to test for statistical significance. *$p \leq 0.05$; ***$p \leq 0.001$; ns, $p \geq 0.05$. PB, peripheral blood; RNP, ribonucleoprotein; UTF, untransfected; MFI, mean fluorescence intensity.

mRNA. Alternatively, delivery of transposable vector elements into NK cells results in multiple potential integration sites, also raising insertional mutagenesis risks.[19,20] None of these approaches allow for a genome edit in which a specific gene can be targeted for knockout (KO) or the precise knockin of a transgene. Non-viral genome editing of NK cells with CRISPR genome editors has been achieved with high efficiency for KOs, but with low efficiency for knockin of large transgene insertions. For example, efficiencies range from 3% to 17% for fluorescent proteins like GFP (1.5-kb insertion) and mCherry (2.1-kb insertion) transgenes, and less than 10% for a CAR (1.4-kb insertion).[21–24]

Here, we describe a robust workflow using non-viral CRISPR-Cas9 ribonucleoproteins (RNP) to generate CAR NK cells against the diasialoganglioside GD2, an antigen expressed on a variety of solid tumors such as neuroblastoma, osteosarcoma, and melanoma.[25] We identify several parameters influencing editing efficiencies to ensure efficient KO of the *KLRC1* gene, encoding the inhibitory NKG2A receptor, and knockin of an anti-GD2 CAR transgene at this locus. *KLRC1*-CAR NK cells are highly viable and proliferative, with minimal off-target editing. In a GD2$^+$ melanoma model, the *KLRC1*-CAR NK cells display improved cytotoxicity and secretion of interferon (IFN)-γ *in vitro* when compared with unedited NK cells. This streamlined, virus-free engineering strategy has high potential to generate allogeneic CAR NK cells for the treatment of HLA-E$^+$ solid tumors.

RESULTS

CRISPR editing of the NKG2A receptor

Our editing strategy to improve NK potency against solid tumors involves knocking out the NKG2A receptor, which inhibits NK cell cytotoxicity upon engagement with the HLA-E ligand on the tumor (Figure 1A), while simultaneously knocking-in a CAR conferring specificity to the GD2 antigen. To knock out the NKG2A receptor, we designed three single-guide RNAs (sgRNAs) against several common coding exons in the human *KLRC1* gene (Figure 1B). Ribonucleoproteins (RNPs) were generated upon the complexation of these sgRNAs with *Sp*Cas9 to generate double-stranded DNA breaks at these targets within *KLRC1*. As a proof of concept, NK-92 cells, an interleukin (IL)-2 dependent human NK cell line, were nucleofected with RNPs containing one of these sgRNAs and assessed for insertions or deletions (indels) around the targeted site after 2 weeks of cell culture. Next-generation sequencing (NGS) of PCR amplicons derived from genomic DNA showed a high level of indel formation (98%) in the RNP-treated cells for our top-performing guide RNA (gRNA1) targeting exon 3 of *KLRC1*. These indels were absent in the untransfected (UTF) control sample (Figure 1C). Moreover, the indels were centered exactly at the predicted site of DNA double-strand cleavage by *Sp*Cas9 (Figure 1D). While UTF NK-92 cells are highly cytotoxic and have been used as an allogeneic cell therapy for cancer for over 20 years, including CAR-modified NK-92s, their cancerous origin requires them to be irradiated before infusion to prevent proliferation, limiting their *in vivo* persistence.[26–30] Additionally, their lack of CD16 expression renders them incapable of antibody-dependent cellular cytotoxicity (ADCC).[31] For these reasons, we then elected to use *ex vivo* expanded primary NK cells isolated from human peripheral blood for downstream development.

NK cells were isolated from the whole blood of healthy donors using density-based negative selection and expanded with irradiated K562-mb15-41BBL feeder cells supplemented with IL-2 and IL-15. After 4 days of culture, NK cells were nucleofected with an RNP complex targeting the *KLRC1* gene and expanded for 1 week. The edited cells were then analyzed for NKG2A expression using flow cytometry alongside a donor-matched UTF control population. Within the RNP-treated cells, an average of 70% NK cells were negative for NKG2A on their cell surface, compared with 13% of NK cells within the UTF control (Figure 1E). The loss of surface NKG2A protein was further corroborated by a decrease in the mean fluorescence intensity (MFI) signal in the RNP-treated cells when compared with the UTF cells (Figure 1F). *KLRC1*-KO NK cells maintained high viability throughout expansion and proliferated consistently (Figures 1G and 1H). The fold expansion of *KLRC1*-KO NK cells was not significantly different from that of their UTF counterparts (Figure 1I).

Non-viral transgene knockin into *KLRC1*

To assess whether this editing strategy could also be used to insert long transgenes in primary NK cells, we adapted a virus-free editing technique developed for primary T cells that harnesses homology-directed repair (HDR) at double-stranded DNA breaks.[32] We nucleofected NK cells with an RNP complex together with a linear, double-stranded DNA (dsDNA) template encoding a long transgene (2,709 bp): the signaling-inert mCherry gene (*KLRC1*-No CAR) (Figure 2A). Since HDR is most likely to occur when cells are actively proliferating and entering the S- or G2-phases of the cell cycle,[33] we first determined the optimal time to nucleofect the cells to ensure the highest knockin efficiencies. We nucleofected NK cells on days 4–7 of expansion and assessed the level of NKG2A knockout and mCherry expression by flow cytometry. mCherry integration was highest when cells were nucleofected on day 4 with an average of 4% cells exhibiting mCherry expression and 68% cells demonstrating NKG2A knockout (Figures 2B and 2C). However, minimal cell growth was observed for the first 6 days after nucleofection (Figure 2D). On days 5, 6, and 7, the percentage of mCherry integration was significantly decreased, to roughly 1% (Figure S1A). In contrast, these later timepoints increased the percentage of NKG2A^{-} cells, with KO rates approaching 80% (Figure S1B).

To increase HDR, the DNA-dependent protein kinase (DNA-PK) inhibitor, M3814, was added to the cell culture medium, as others had previously shown it to improve knockin rates of primary human T cells by inhibiting non-homologous end-joining (NHEJ) DNA repair events.[22] NK cells were nucleofected on day 4 with the RNP complex and mCherry dsDNA, and incubated with M3814 at various concentrations. Nucleofected cells were expanded for 1 week and assayed by flow cytometry for NKG2A KO and transgene expression. Incubation with M3814 had no effect on NKG2A KO efficiencies, with the percent of NKG2A^{-} cells remaining around 75% (Figure S1C). In contrast, knockin efficiencies significantly improved when nucleofected cells were treated with any concentration of M3814 compared with no M3814 treatment (Figure 2E). Cells treated with either 0.5 μM or 1 μM of M3814 had the highest percentage of transgene expression at 11.9% and 12.8%, respectively. Moreover, treatment with M3814 did not affect cell expansion after nucleofection, with M3814-treated cells growing comparably to untreated cells (Figure 2F). For subsequent studies, 0.5 μM M3814 was chosen due to the increased editing efficiency and high cell yield at 1 week after nucleofection.

As our strategy depends on electroporation for the efficient delivery of payload into NK cells, we next optimized the pulse codes to strike a balance between knockin efficiency and cell viability. For the Lonza 4D nucleofection system, we selected two programs (DN-100 and EH-100) that have resulted in high RNP-based KO efficiencies and plasmid-based GFP knockin efficiencies while maintaining cell viability in primary NK cells.[21] Additionally, we included two pulse codes (DP-100 and ER-100) that were recommended by the manufacturer as being similar to DN-100 but potentially yielding higher efficiencies. As a control, we also tested the EH-115 program used to nucleofect primary human T cells.[32,34] Consistent with the M3814 optimization, there were no significant differences in NKG2A KO efficiencies across the different programs, with the percent of NKG2A^{-} cells remaining between 76% and 85% (Figure S1D). The EH-100 program yielded the highest percentage of transgene-positive NK cells at 23.5% (Figure 2G). Finally, we assessed cell viability and growth throughout expansion for each of the pulse code programs to maximize cell yield. Throughout expansion, the DN-100 program resulted in higher NK cell yields, while the ER-100 program yielded the lowest number of viable NK cells (Figure 2H). The EH-100 pulse code was chosen for subsequent experiments due to the significant improvement in transgene editing and sufficient expansion of the edited NK cells.

Characterization of *KLRC1*-CAR NK cells

Having developed an efficient non-viral *KLRC1* gene editing protocol for NK cells (Figure 3A), we applied our strategy to generate

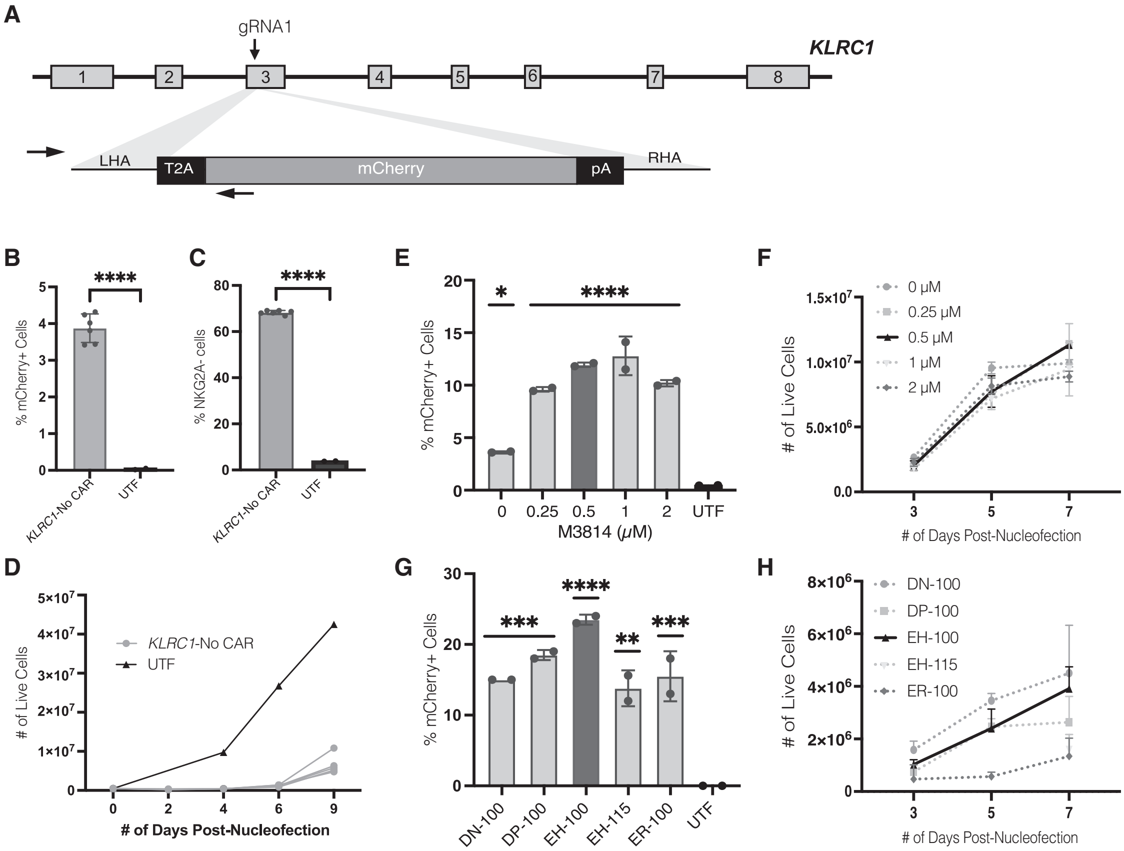
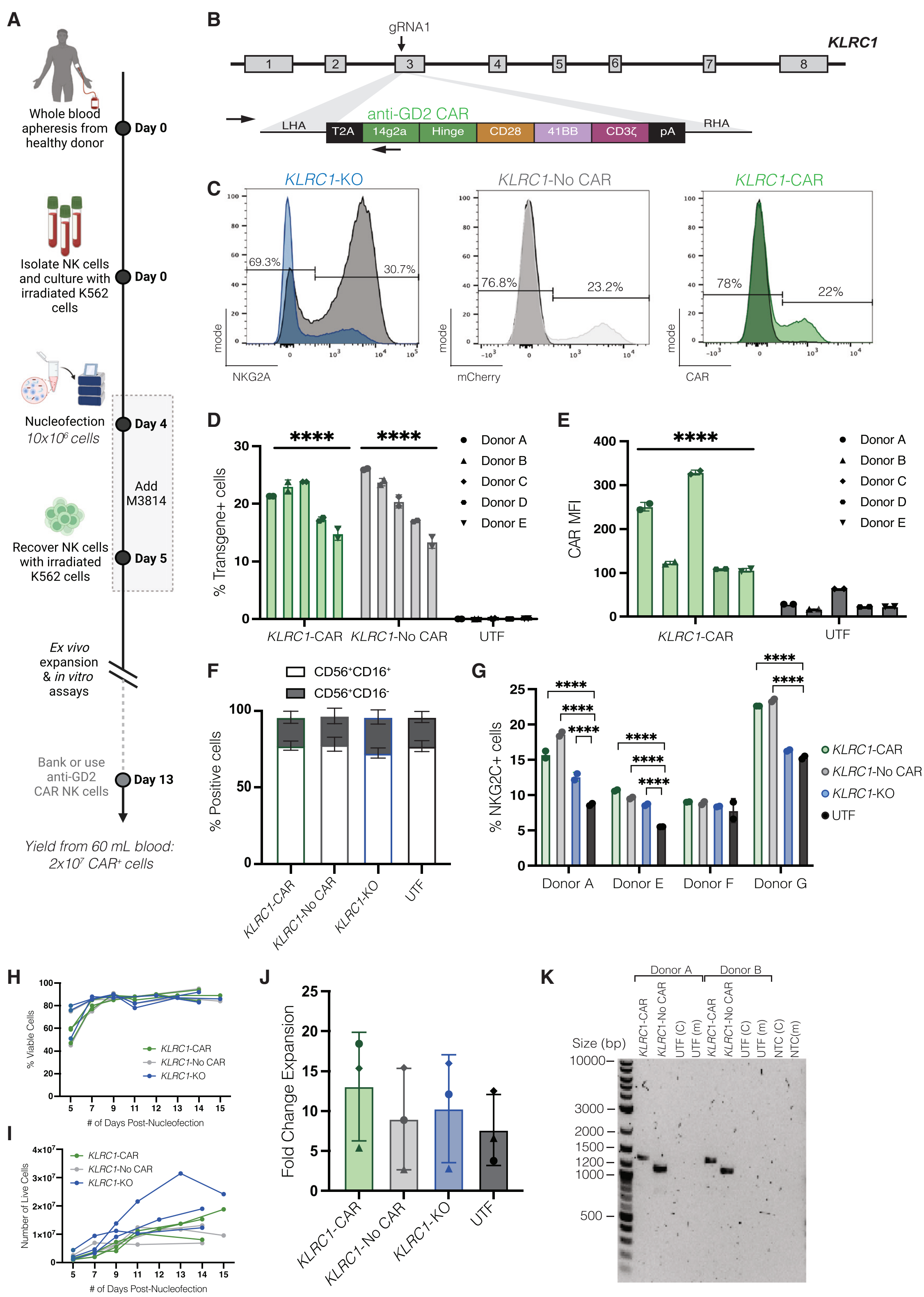


Figure 2. Non-viral transgene knockin into primary NK cells

See also Figure S1. (A) Schematic depicting virus-free insertion of the mCherry fluorescence gene at the third exon of the *KLRC1* locus. (B and C) NK cells were nucleofected on day 4 of expansion. Percentage of mCherry$^+$ cells and NKG2A$^-$ cells is shown, as measured by flow cytometry, 1 week after nucleofection. Data are shown for six replicates for one donor (*n* = 1). (D) Cell proliferation of *KLRC1*-No CAR cells from B. (E) NK cells were incubated with varying concentrations of M3814 for 24 h after nucleofection. The percent of mCherry$^+$ cells was measured by flow cytometry 1 week after nucleofection. Data are shown for two replicates for one donor (*n* = 1). (F) Expansion of mCherry$^+$ cells after a 24-h incubation with M3814. (G) NK cells were nucleofected on day 4 using different pulse programs, followed by M3814 treatment. Data are shown for two replicates for one donor (*n* = 1). (H) Expansion of NK cells following nucleofection with different pulse programs. Statistical significance was calculated using a two-tailed unpaired t test (B and C) and by ordinary one-way ANOVA. Dunnett's multiple comparison test was used as the post-test (E and G). *$p \leq 0.05$; **$p \leq 0.01$; ***$p \leq 0.001$; ****$p \leq 0.0001$. ANOVA, analysis of variance.

GD2-targeting CAR NK cells (*KLRC1*-CAR) with an anti-GD2 CAR composed of the following: an anti-GD2 single chain variable fragment (scFv; 14g2a), CD28 transmembrane and endodomain, 41BB costimulatory domain and CD3z cytotoxicity domain (Figure 3B). Though the CAR transgene (3,180 bp) was approximately 500 bp larger than the mCherry transgene (2709 bp), the CAR knockin efficiency (average of 20.0%) was similar to that of mCherry (average of 20.1%) (Figures 3C and 3D). The MFI for the CAR transgene also significantly increased compared with the donor-matched UTF control (Figure 3E). In line with previous outcomes, the NKG2A KO efficiencies remained largely unchanged when NK cells received the CAR transgene but were increased marginally with the delivery of the mCherry transgene (Figure S2A). The NKG2A MFI levels, however, had no significant differences across the three groups that were edited via CRISPR (Figure S2B). After profiling for canonical CD56 and CD16 NK cell surface markers in the cell products, the fractions of CD56$^+$CD16$^+$ and CD56$^+$CD16$^-$ cells among the edited and untransfected groups remained unchanged through the editing and expansion workflows (Figure 3F). In contrast, a donor-dependent increase in NKG2C expression was observed among the edited groups in comparison to UTF controls, with *KLRC1*-CAR and *KLRC1*-No CAR cells harboring the greatest expression (Figure 3G).

We next confirmed that insertion of the CAR into the *KLRC1* locus did not affect NK cell viability and proliferation by tracking cell expansion after nucleofection. When comparing the various cell products immediately after nucleofection, poor cell expansion correlated with the length of the exogenous dsDNA template within the first week, with *KLRC1*-CAR NK cells having the lowest fold expansion, followed by *KLRC1*-No CAR, and *KLRC1*-KO NK cells (Figures S2C–S2E). However, after recovery from electroporation, *KLRC1*-CAR NK cells were highly viable and proliferated at a similar



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rate to *KLRC1*-No CAR and *KLRC1*-KO NK cells in the log growth phase (Figures 3H and 3I). The fold change expansion for all edited groups beginning on day 5 was comparable to that of donor-matched UTF NK cells with no significant differences (Figure 3J).

We next characterized the genomic outcomes within the *KLRC1*-CAR NK cell product by checking for on- and off-target modifications, and integration of our transgenes throughout the genome. First, an "in-out" PCR for the intended transgene integration at the *KLRC1* on-target site was performed on genomic DNA extracted from edited NK cells. Using primers that were specific to the transgene and the *KLRC1* locus, we showed site-specific amplification of our GD2 CAR and mCherry transgenes that were not present in the UTF cells (Figure 3K). The size of the bands are approximately 1,300 bases (CAR) and 1,100 bases (mCherry), exactly as predicted from the proper on-target integration of each transgene (Table S1).

We also examined the genome for off-target activity from the *KLRC1* gRNA via CHANGE-seq. This method profiles the entire human genome for dsDNA breaks induced by a Cas9 RNP. Moderate off-target activity in the CHANGE-seq assay was observed largely within intergenic or intron regions of the genome, with 10 sites having off-target nuclease activity within exons and eight sites having off-target activity near a transcription start site (TSS) (Figure S3A; Table S2). The CHANGE-seq specificity ratio, calculated by dividing the number of on-target reads by the sum of total reads, for our sgRNA is 0.038. As expected, small mismatches distal to the protospacer-adjacent motif (PAM) of Cas9 were present in the top-nominated off-target sites (Figure 4A).

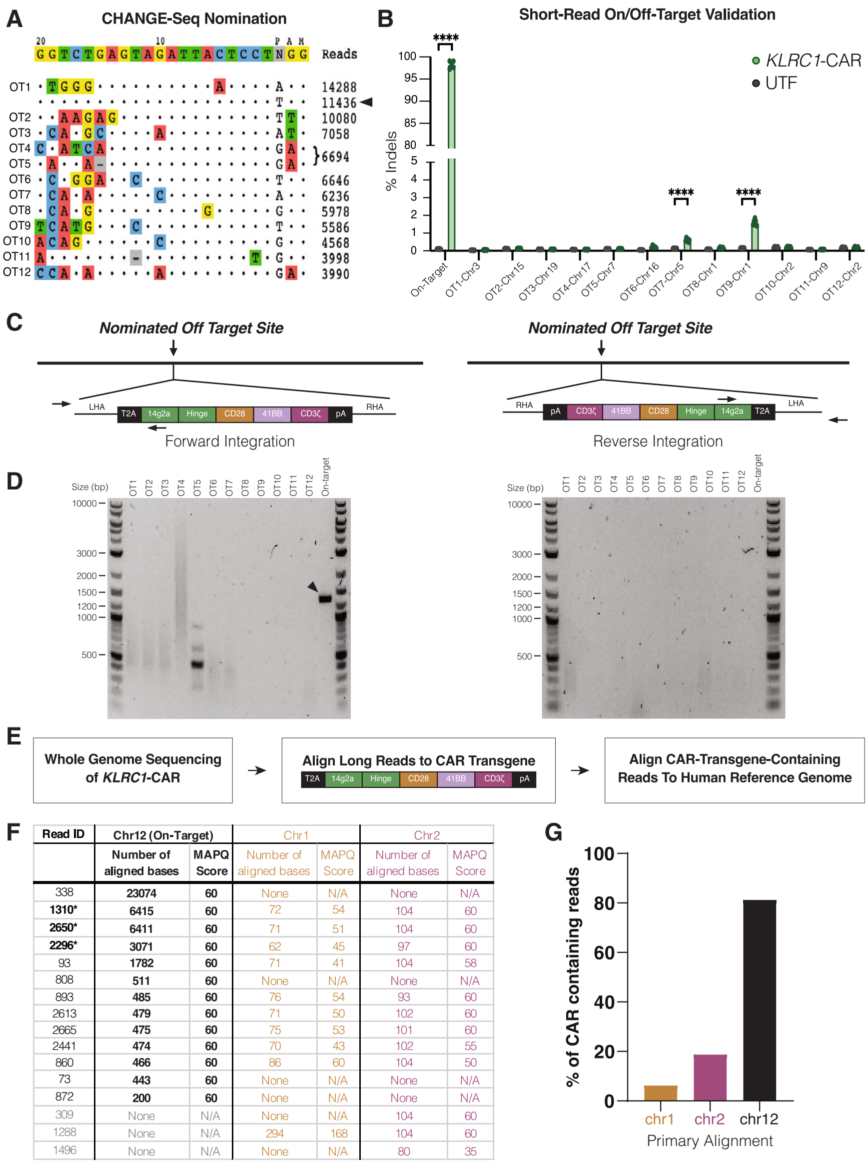
To further evaluate potential off-target genomic modifications, we performed amplicon sequencing around each of the top 12 off-target sites nominated by CHANGE-seq (>3,500 reads) within *KLRC1*-CAR and UTF NK cells. We completed NGS via the rhAmpSeq analysis system to multiplex the genomic analysis for on- and off-target sites from the same genomic DNA sample. By rhAmpSeq, we observed high indel formation only at the on-target site with low or undetectable levels of editing across all 12 off-target sites (Figure 4B). At two sites, modest but significant increase in off-target activity was observed compared with UTF controls.

To evaluate the potential off-target integration of our CAR, we performed in-out PCR on each of the top 12 off-target sites across three donors to check for unintended CAR integration, in either the 5′-3′ or 3′-5′ orientation (Figure 4C). No bands of the expected size were observed (Table S1) in any of the off-target sites, indicating that there is no off-target integration of the CAR within the limit of detection for this assay (Figures 4D and S3B). The band at ∼500 bp at site 5 (Figure 4D, left, lane 6) is from non-specific binding of the primers as the expected amplicon size from off-target integration at this site would have been 1,931 bp. To confirm that the band observed at ∼500 bp was from non-specific binding of the primers, PCR analysis of site 5 was repeated alongside donor-matched UTF controls. Amplification of the ∼500 bp band was observed within both *KLRC1*-CAR and UTF groups (Figure S3C), confirming non-specific primer activity. As a positive control, the on-target site was included, and a band of approximately 1,300 bp was observed (Figures 4D and S3B, left, lane 14), which corresponds to the expected amplicon size of 1,349 bp, confirming successful on-target integration of the CAR within *KLRC1*.

We performed long-read whole-genome sequencing (WGS) of *KLRC1*-CAR NK cells to track integration sites of our transgene unbiasedly, as WGS does not have amplification biases from PCR nor does it require nomination of off-target sites. Genomic DNA was prepared for long-read sequencing via Oxford Nanopore Technologies (ONT) and aligned to a reference sequence containing the CAR transgene (Figure 4E). Successful mapping to the reference human

Figure 3. Characterization of non-viral *KLRC1*-CAR NK cells

See also Figure S2. (A) Proposed manufacturing timeline for *KLRC1*-CAR NK cells. NK cells are isolated from whole blood of healthy donors using negative selection and activated with irradiated K562-mb15-41BBL cells (1 NK: 2 K562) along with IL-2 and IL-15. On day 4, NK cells are nucleofected with Cas9 RNP and dsDNA HDRT using program EH-100, followed by a 24-h treatment of 0.5 μM M3814. Nucleofected NK cells are recovered with addition of irradiated K562-mb15-41BBL cells (1:2), supplemented with IL-2 and IL-15. *KLRC1*-CAR cells are expanded *ex vivo* for downstream analysis and cell banking. The calculated cell yield reflects the estimated number of *KLRC1*-CAR$^+$ NK cells derived from 60 mL of blood on day 13 of expansion. (B) Schematic depicting virus-free insertion of an anti-GD2 CAR at the third exon of the *KLRC1* locus. (C) Representative histogram plots showing knockout of NKG2A and knockin of CAR or mCherry transgenes 1 week after nucleofection. The x axis describes protein expression levels; the y axis describes frequency after normalization to the mode. UTF samples are shown as black histograms in all plots. The percent shown for each gate represents protein levels in the *KLRC1*-KO (left), *KLRC1*-No CAR (middle), and *KLRC1*-CAR (right) edited samples. (D and E) CAR and mCherry knockin efficiencies, and MFI levels are shown. Data are shown as an average of two replicates across five donors (n = 5). (F) The percent of CD56$^+$CD16$^+$ and CD56$^+$CD16$^-$ cells in each cell population is shown. Data are shown as an average of two replicates across two donors (n = 2). (G) The percent of NKG2C+ cells in each population is shown. Data are shown as an average of two replicates for each of the four donors (n = 4). (H and I) Cell viability and cell expansion of *KLRC1*-CAR, *KLRC1*-No CAR, and *KLRC1*-KO NK cells after nucleofection are shown. During expansion, multiple nucleofection reactions were pooled to form one sample per group. Data are normalized to the number of reactions for each of the three donors (n = 3). (J) Calculated fold change expansion of *KLRC1*-CAR, *KLRC1*-No CAR, and *KLRC1*-KO NK cells from (H) is shown in comparison to donor-matched UTF cells, with each donor indicated by a different shape (circle, triangle, diamond). (K) In-out PCR showing on-target integration of the CAR and mCherry transgenes across two donors. The UTF samples were amplified using the CAR (UTF [C]) and mCherry (UTF [m]) primer pairs, as controls. NTC refers to the PCR reaction performed without the genomic DNA. The primers are indicated by the black arrows on each template in the schematics of Figures 2A and 3B. Statistical significance was calculated using an ordinary one-way or two-way ANOVA. Post-tests were performed using the uncorrected Fisher's least significant difference (D), Sidak's multiple comparisons test (E), and Dunnett's multiple comparisons test (G and J). *$p \leq 0.05$; ****$p \leq 0.0001$; ns, $p \geq 0.05$. ANOVA, analysis of variance; CAR, chimeric antigen receptor; dsDNA, double-stranded DNA; HDRT, homology-directed repair template; MFI, mean fluorescence intensity; NTC, non-template control; PCR, polymerase chain reaction; RNP, ribonucleoprotein; UTF, untransfected.



Read ID	Chr12 (On-Target)		Chr1		Chr2	
	Number of aligned bases	MAPQ Score	Number of aligned bases	MAPQ Score	Number of aligned bases	MAPQ Score
338	23074	60	None	N/A	None	N/A
1310*	6415	60	72	54	104	60
2650*	6411	60	71	51	104	60
2296*	3071	60	62	45	97	60
93	1782	60	71	41	104	58
808	511	60	None	N/A	None	N/A
893	485	60	76	54	93	60
2613	479	60	71	50	102	60
2665	475	60	75	53	101	60
2441	474	60	70	43	102	55
860	466	60	86	60	104	50
73	443	60	None	N/A	None	N/A
872	200	60	None	N/A	None	N/A
309	None	N/A	None	N/A	104	60
1288	None	N/A	294	168	104	60
1496	None	N/A	None	N/A	80	35

Figure 4. Genome-wide off-target analysis

See also Figure S3. (A) Nominated off-target sites detected by CHANGE-seq are organized by number of read counts. The on-target site is the second line on the list, indicated by a black arrow. Base pair mismatches between the sgRNA and the off-target sites are shown as colored nucleotides ($n = 1$). (B) Editing across the top 12 nominated off-target sites was assayed via amplicon using the rhAmpSeq system. Data are shown as mean (SD) across three donors, with two replicates per donor ($n = 3$). (C) Schematic depicting locations of the forward and reverse primers for in-out PCRs for the CAR gene at an off-target site. The left schematic represents CAR integration in the 5′–3′ orientation, while the right represents integration in the 3′–5′ orientation, with the primers indicated by black arrows. (D) In-out PCRs were performed on genomic DNA extracted from *KLRC1*-CAR NK cells to assess for off-target integration in the forward (left) and reverse (right) orientations. The on-target CAR integration is indicated by the black arrow. (E) Schematic of the methods used to analyze long-read WGS data. Following WGS, reads containing the CAR transgene were aligned to the human genome and filtered by map quality and overlap window. (F) Alignment statistics for each of the 16 reads containing the CAR transgene. The reads marked by an asterisk indicate contiguous mapping of the CAR construct several thousand base pairs upstream of the LHA and downstream of the RHA. (G) The percentage of all 16 reads aligning to each of the identified chromosomal regions. A two-way ANOVA was used to calculate statistical significance, with an uncorrected Fisher's least significant difference post-test (B). CAR, chimeric antigen receptor; PCR, polymerase chain reaction.

genome provided 30X coverage of the genome. On-target integration was observed in 13 of 16 reads with alignment to the CAR transgene. An exact match to the predicted HDR product at the intended on-target *KLRC1* locus on Chromosome 12 was observable in several distinct long reads spanning the insertion and homology arms (Figures 4F and 4G). These single reads had the highest possible score (MAPQ of 60) and showed contiguous mapping to Chromosome 12 as far as 7,000 bases upstream and 4,000 bases downstream of the arms (Figures 4F and 4G). Of the reads containing the CAR transgene, 81% of them mapped to portions of the on-target site with high scores within the specified overlap window (Figure 4G). The remaining 19% of the reads mapped partially to the CAR and Chromosomes 1 and 2. However, the alignment length of these reads is quite small compared with the on-target alignments: on average, only 71 bp alignment for Chromosome 1 and 101 bp for Chromosome 2 vs. 1,362 bp for Chromosome 12 (Figure 4F). Upon closer examination, the alignment of the reads on Chromosomes 1 and 2 mapped to CAR domains that are present in the native human genome, such as CD28 on Chromosome 1 and CD3-zeta on Chromosome 2. Overall, WGS results confirmed high levels of precise on-target integration from our genome editing strategy.

KLRC1-CAR NK cells are highly cytotoxic against GD2$^+$ tumor cells

The specificity of *KLRC1*-CAR NK cell function was evaluated against GD2$^+$ M21 melanoma cells (Figure 5A). We isolated pure populations of NKG2A$^-$, CAR$^+$, or mCherry$^+$ cells using florescence activated cell sorting (FACS) to determine the effect of each gene modification on NK cytotoxicity. NK cells were co-cultured with M21 cells at a 5:1 effector-target ratio for 24 h (Figure 5B). NK cytotoxicity was 6%–10% higher in the *KLRC1*-CAR NK cells compared with *KLRC1*-No CAR, *KLRC1*-KO, and UTF cells (Figure 5C, left). A monoclonal antibody to label the CAR target antigen, GD2, was added to the

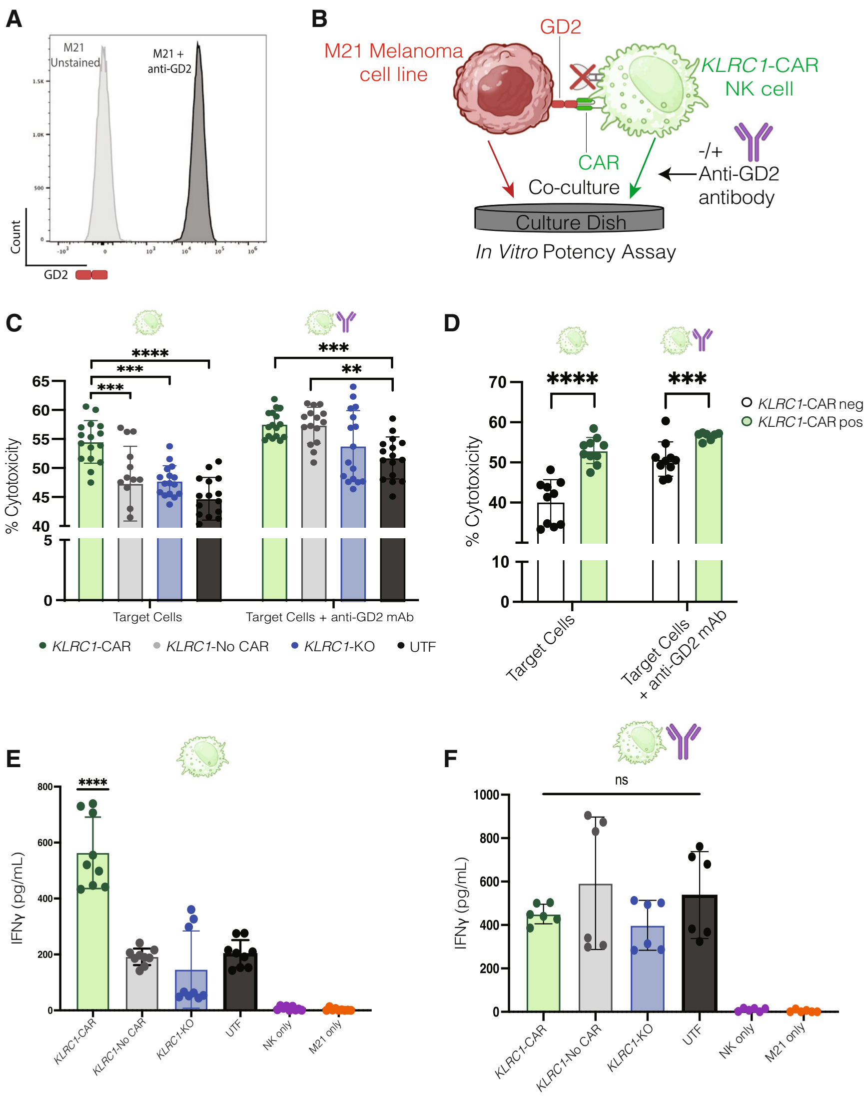


Figure 5. *KLRC1*-CAR cells display improved cytotoxicity against GD2+ melanoma cells *in vitro*

See also Figure S4. (A) Flow cytometry histogram plot depicting GD2 expression on the surface of M21 melanoma cells. (B) Schematic depicting the *in vitro* potency assay used to assess NK cell cytotoxicity. (C) Following FACS, NK cells were co-cultured with M21 cells at a 5:1 E:T ratio for 24 h. For ADCC conditions, post-FACS NK cells were co-cultured with M21 cells in the presence of 500 ng/mL hu14.18K322A anti-GD2 monoclonal antibody. The non-ADCC conditions are indicated with a single green NK cell icon above the graph, while the ADCC conditions are indicated by cell and antibody icons. Cytotoxicity was measured by quantifying secreted LDH with the CyQuant LDH Cytotoxicity Assay kit. Data are shown as the mean (SD) across three donors, with 5–6 replicates per donor (n = 3). (D) Cytotoxicity was measured by the same LDH release assay as in (C), but for CAR^+ and CAR^- fractions within the same sample. Data are shown as the mean (SD) across three donors, with 5–6 replicates per donor (n = 3). (E and F) NK cell-secreted IFNγ was measured using an ELISA on media supernatant taken from the 24-h co-culture with M21 cells for the target cells only (E) and the ADCC (F) conditions. Data are shown as mean (SD) for three donors with three replicates (n = 3) for the target cells only (E) and two donors with three replicates (n = 2) for the ADCC (F) conditions. Statistical significance was calculated using an ordinary two-way ANOVA, with the Dunnett's multiple comparison test (C), and the Sidak's multiple comparison post-test (D), and a one-way ANOVA with the Tukey's multiple comparison post-test (E and F). *$p \leq 0.05$; **$p \leq 0.01$; ***$p \leq 0.001$; ****$p \leq 0.0001$; ns, $p \geq 0.05$.ANOVA, analysis of variance; ADCC, antibody-dependent cellular cytotoxicity; CAR, chimeric antigen receptor; LDH, lactate dehydrogenase.

co-culture so that NK cell recognition of the tumor could occur independent of the CAR via CD16, triggering ADCC. Upon addition of the hu14.18K322A anti-GD2 monoclonal antibody, all NK cell products were highly cytotoxic (Figure 5C, right) and the differences in NK cell cytotoxicity among *KLRC1*-CAR, *KLRC1*-No CAR, and *KLRC1*-KO NK cells were erased in ADCC-promoting conditions. Upon comparing non-ADCC and ADCC conditions, we attribute that the boost in the cytotoxic activity of the *KLRC1*-CAR NK cells compared with other cells is specific to the GD2 antigen. The precise knockin of the CAR increases the cytotoxicity of the *KLRC1*-CAR NK cells to the maximal level in this co-culture assay.

To assess whether the improvement in cytotoxicity within the *KLRC1*-CAR NK cells was observed across other tumor models, we cultured the engineered NK cell products with GFP-labeled, GD2-expressing CHLA-20 neuroblastoma cells and measured NK cell killing. At the 1:1 E:T, *KLRC1*-CAR^+ NK cells displayed the highest cytotoxicity compared with *KLRC1*-CAR^- and *KLRC1*-No CAR at the 8-, 12-, and 24-h time points (Figure S4). As the number of NK cells was increased to reach an E:T of 5:1, the differences between *KLRC1*-CAR cells and other NK cell populations were minimized within the 24-h window.

To further verify that the increase in cytotoxicity is due to the CAR, we isolated CAR^+ and CAR^- cell fractions within a sample and measured NK cytotoxicity. The CAR^+ fractions exhibited 13% greater cytotoxicity than their CAR^- counterparts (Figure 5D). Again, this difference between the two populations was completely erased upon addition of the anti-GD2 monoclonal antibody under ADCC conditions.

Analysis of media supernatant indicated that the *KLRC1*-CAR NK cells secreted the greatest IFNγ upon antigen exposure (Figure 5E). In agreement with the cytotoxicity data, there were no significant differences in IFNγ production among the *KLRC1*-No CAR, *KLRC1*-

KO, and UTF groups. Under ADCC conditions, all four groups produced an equivalent amount of IFNγ after antigen exposure (Figure 5F). Interestingly, all groups except the *KLRC1*-CAR NK cells secreted a greater total amount of IFNγ when compared with their non-ADCC counterparts.

KLRC1-CAR NK cells overcome HLA-E-based inhibition

To interrogate the combinatorial effect of the NKG2A KO and CAR insertion, we assessed the potency of the *KLRC1*-CAR NK cells against HLA-E expressing M21 cells. Following transduction with an HLA-E vector containing an HLA-G leader peptide,[10] 44% of M21 cells expressed HLA-E on the cell surface, with expression increasing to 100% after a 24-h treatment with 100 ng/mL of IFNγ (Figure 6A). Pure populations of NKG2A^{-}, CAR^{+}, or mCherry^{+} cells were isolated using FACS and co-cultured with IFNγ-treated M21-HLA-E cells at various effector-target ratios for 24 h alongside donor-matched UTF controls (Figures 6B–6D and S5A–S5C). At the lowest ratio (0.1:1), there were no differences in NK-mediated cytotoxicity between the groups. When cultured at the 0.5:1 ratio, *KLRC1*-CAR NK cells displayed superior cytotoxicity compared with *KLRC1*-No CAR, *KLRC1*-KO, and UTF cells. At the highest ratio, *KLRC1*-CAR NK cells continued to exhibit improved cytotoxicity when compared with UTF cells by 17%, respectively, but no significant differences were observed relative to the *KLRC1*-No CAR or *KLRC1*-KO NK cells. Again, in ADCC conditions, the differences in cytotoxicity were nullified between the edited groups at all three effector-target ratios, mirroring the trends seen against the wild-type M21 cells (Figure S5D). Finally, we confirmed that the improvement in NK potency against HLA-E-expressing cancer cells was still driven by CAR activity by analyzing the cytotoxicity of CAR^{+} and CAR^{-} fractions from within the same sample (Figure 6E). The CAR^{+} fractions displayed improved cytotoxicity at each of the effector:target ratios by 2%, 11%, and 18%, respectively. Under ADCC conditions, these differences were minimized at the 0.5:1 and 1:1 ratios to 5% and 7%, respectively, with a slight increase at the 0.1:1 ratio to 3.5%.

KLRC1-CAR NK cells exhibit greater killing per cell

To track *KLRC1*-CAR NK cell cytotoxicity at a single-cell resolution, we imaged the dynamics between the NK cells and the M21-HLA-E cells through time-lapse imaging (Figure 7A). *KLRC1*-CAR NK or UTF NK cells were co-cultured with M21-HLA-E cells and were imaged for 16 h (Figures 7B and 7C). An increase in apoptotic fluorescence intensity was observed over the course of the co-culture within both effector groups, with the *KLRC1*-CAR NK cells inducing a faster rate of increase, suggesting a quicker and more potent response (Figure 7D). When comparing the final apoptotic fluorescence intensity values for each effector group at 1 and 16 h, *KLRC1*-CAR NK cells displayed a greater increase in signal intensity than their UTF counterparts (Figure 7E). For analysis at the single-cell level, 15 representative NK cells were chosen per condition and tracked for the number of M21-HLA-E cells they killed. A higher percentage of *KLRC1*-CAR NK cells displayed cytotoxic capacity against the target cells when compared with donor-matched UTF controls (Figure 7F). Moreover, *KLRC1*-CAR NK cells also demonstrated the ability to kill multiple M21-HLA-E cells simultaneously, unlike UTF controls, suggesting that the improvement in potency could partly be attributed to “burst killing” (Figures 7G and 7H). Importantly, the *KLRC1*-CAR NK cells display better target killing per NK cell tracked, indicating the genome editing strategy produces potent functional responses against HLA-E-expressing target cells at the single-cell level.

DISCUSSION

NK cells often fail to control solid tumors due to the immunosuppressive conditions of the tumor microenvironment, which can include the expression of ligands (e.g., HLA-E) that engage inhibitory receptors on NK cells like NKG2A.[35] Though NK cells have previously been genetically modified to successfully improve cytotoxicity and persistence using viral vectors, transposons, mRNA transfection, and CRISPR+AAV technologies, these methods elicit concerns of safety, lack the machinery for stable editing, or have knockin limitations on transgene size.[19,20,36–39] Here, we demonstrate an optimized CRISPR-Cas9-enabled approach that generates stable and precisely edited primary NK cells without the use of viral vectors. Our data show that the CRISPR platform can be leveraged for efficient KO of the inhibitory NKG2A receptor and virus-free knockin of an anti-GD2 CAR. We illustrate a high knockin efficiency within primary NK cells for large transgenes (∼3.1 kb) with little to no off-target editing. Moreover, we also validate that *KLRC1*-CAR NK cells are viable, proliferative, and capable of inducing tumor lysis and cytokine secretion *in vitro.* Taken together, our work establishes that *ex vivo* expanded NK cells from peripheral blood can be effectively modified to express a functional CAR, while also deleting an immune checkpoint molecule using CRISPR genome editing.

Despite nominating many off-target sites for the sgRNA from the highly sensitive CHANGE-seq assay, we saw minimal indel formation across the off-target sites. One explanation is that the RNP complex is only present in the NK cells transiently, resulting in lower exposure to nuclease activity when compared with the *in vitro* assay conditions used in CHANGE-seq. Furthermore, M3814 may inhibit NHEJ events in cells that experienced an off-target double-stranded break resulting in either cell death or precise HDR. Notably, these findings are similar to those of the recently approved exagamglogene autotemcel (exa-cel, Casgevy) Cas9-edited cell therapy, in which none of the nominated sites were seen upon validation in the cell product.[40,41] The negligible indel formations at sites 7 and 9 within the *KLRC1*-CAR NK cells were found to be in the *HAND1* and *AKNAD1* genes, respectively, neither of which are expressed by NK cells (Human Protein Atlas: Immune Cell Section Summary).[42–45] While the gRNA utilized in this study has a promising on-/off-target profile, it is possible to decrease the off-target activity further by choosing other guides or utilizing nucleases other than *Sp*Cas9. Importantly, little to no off-target integration was observed using the assays described in this study, indicating that the endogenous *KLRC1* promoter is likely the sole driver of CAR expression of our promoter-less CAR transgene. Though virally engineered CAR NK cells have not been associated

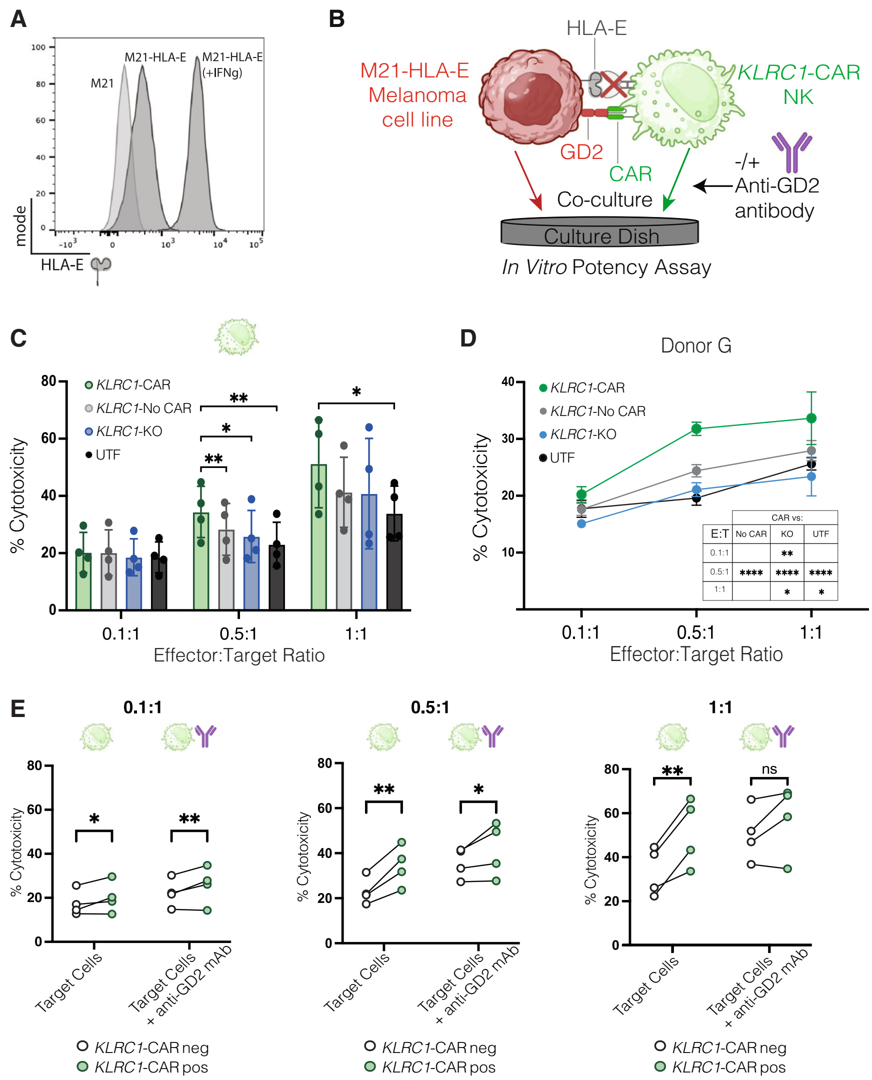


Figure 6. Overcoming the NKG2A-HLA-E-based checkpoint

See also Figure S5. (A) Flow cytometry histogram plot showing HLA-E expression on wild-type M21 cells, HLA-E transduced M21 cells, and HLA-E transduced M21 cells treated with 100 ng/mL of IFNγ for 24 h. (B) Schematic depicting the *in vitro* potency assay used to assess NK cell cytotoxicity against the M21-HLA-E cells. (C) After FACS, NK cells were co-cultured with M21-HLA-E cells at 0.1:1, 0.5:1, and 1:1 effector-target ratios for 24 h. Cytotoxicity was measured by quantifying secreted LDH with the CyQuant LDH Cytotoxicity Assay kit. Data are shown as the mean (SD) of five replicates for four donors (n = 4). (D) Potency data are shown for one representative donor from (C) at 0.1:1, 0.5:1, and 1:1 E:T ratios. Data are shown as the mean (SD) of five replicates. (E) *KLRC1*-CAR NK samples were sorted into CAR$^-$ and CAR$^+$ fractions and were cultured with M21-HLA-E cells under the same conditions as in (C) and (D). The secreted LDH was used to quantify NK cytotoxicity after 24 h. Data are shown as the mean (SD) of five replicates for four donors (n = 4). Statistical significance was calculated using an ordinary two-way ANOVA, with the Dunnett's multiple comparison test (C and D), and a two-way ANOVA with the Sidak's multiple comparison post-test (E). *$p \leq 0.05$; **$p \leq 0.01$; ***$p \leq 0.001$; ****$p \leq 0.0001$; ns, $p \geq 0.05$. ADCC, antibody-dependent cellular cytotoxicity; ANOVA, analysis of variance; CAR, chimeric antigen receptor; LDH, lactate dehydrogenase.

with NK cell lymphomas in patients, the CRISPR strategy described in this study increases the safety profile of NK cell therapies by lowering random transgene integrations, thus establishing a more precise and controlled therapeutic.

Our RNP-based KO strategy yielded high numbers of *KLRC1*$^{-/-}$ cells that were viable and proliferative, at efficiencies higher than those previously reported.[12,13,21] Bexte et al. reported a KO efficiency of 43.5%, while Mac Donald et al. reported an efficiency of 41% for sgRNAs targeting the third exon. Interestingly, Bexte et al. reported a lower indel frequency for the sgRNA used in this study (~58%–80%) in primary T cells. However, protein expression was not reported in that study, which makes it difficult to conclude how the editing translates to KO efficiencies in their system. In contrast, Huang et al. reported high indel frequencies (>80%) for the sgRNA used in this study but showed less than 25% NKG2A KO. Though Mac Donald et al. deliver a greater amount of nuclease, their system uses the Cas12a enzyme, highlighting another difference between the studies. Further optimization of the gRNA design, such as opting for sequences with a lower GC content and a lower hybridization free energy change (−64.53 to −47.09),[46] can help improve the KO efficiencies beyond those observed in each of the studies.

The dsDNA template strategy used in this study for transgene integration has been utilized in primary NK cells before, but for smaller sequences. The studies from the Lin and Marson groups report knockin efficiencies ranging from 3.09%–7.5% for a dsDNA GFP template (~1.5 kb) across several loci.[21–23] One study by Kath et al. reports knockin outcomes of 15% for the GFP gene but less than 10% for a truncated CAR of approximately 1.4 kb in size.[24] Though the knockin efficiencies observed in our study are lower than the efficiencies reported in the studies described above without the use of small molecules, we provide compelling evidence that using HDR enhancers can improve editing outcomes. Nonetheless, a thorough comparison of the nucleofection protocols would provide more insight into which parameters are most optimal for NK cells.

Since HDR is more likely to occur within cells in the S- or -G2 phases,[33] the best knockin percentages occurred when NK cells were expanded with K562-mb15-41BBL feeder cells, IL-2, and IL-15 for

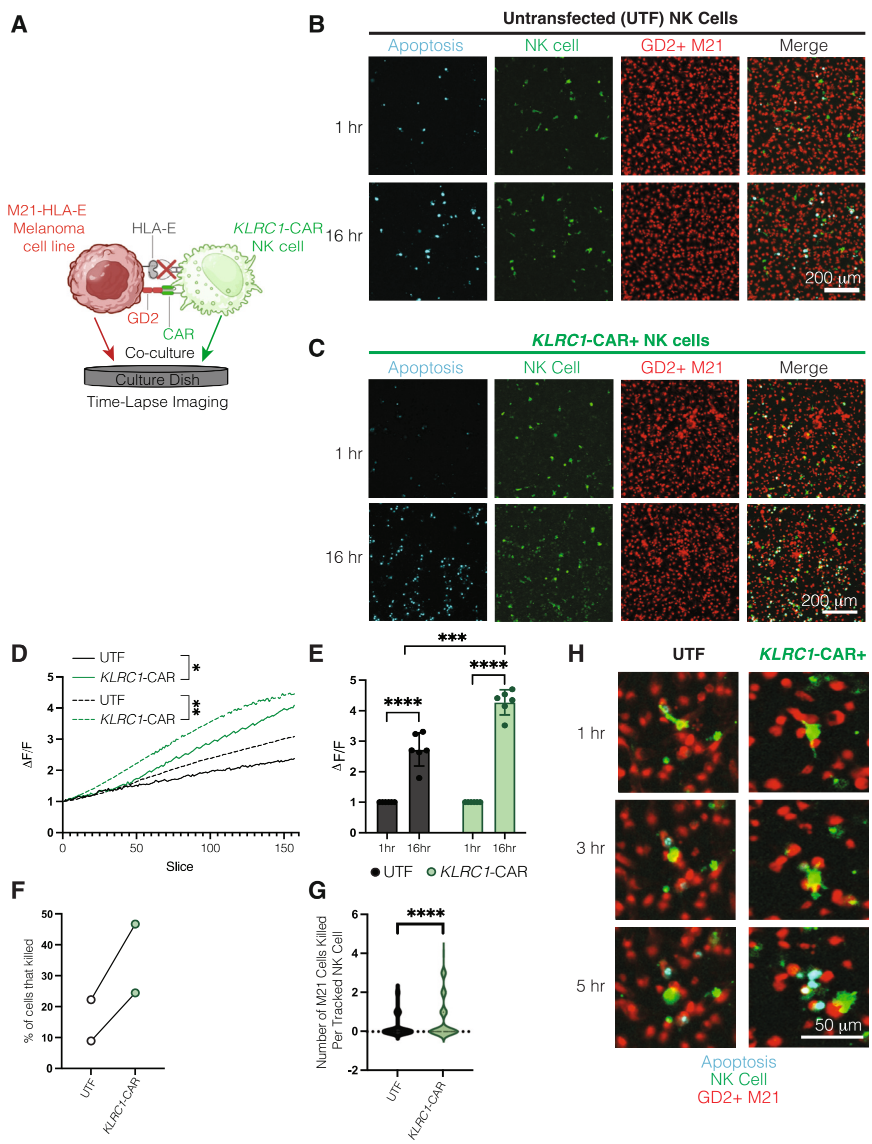


Figure 7. *KLRC1*-CAR NK cells demonstrate improved killing of HLA-E/GD2+ target cells at the single-cell level

(A) Schematic describing the co-culture setup for time-lapse imaging. (B and C) Representative images of time-lapse microscopy of UTF (B) and *KLRC1*-CAR NK (C) NK cells over 16 h. The M21-HLA-E cells are represented in red, the NK cells in green, and apoptosis events in cyan. Scale bar denotes 200-μm length. (D) The change in fluorescence intensity over 16 h for *KLRC1*-CAR NK and UTF NK cells. Donor 1 is represented with a solid line and donor 2 by the dashed line, with *KLRC1*-CAR NK cells in green, and UTF NK cells in black. Data are shown as the mean (SD) of three replicates for each donor ($n = 2$). (E) The difference in fluorescence intensity between 0 and 16 h was quantified for both effector groups. Data are shown as the mean (SD) of three replicates for two donors ($n = 2$). (F) The percentage of NK cells within each effector group that killed at least one M21-HLA-E target cell. Data are shown as the mean of three replicates for two donors ($n = 2$). (G) The number of M21-HLA-E target cells killed by an individual NK cell. Data are shown as the mean (SD) of three replicates for two donors ($n = 2$). (H) Representative image of UTF NK cells and *KLRC1*-CAR NK cells lysing M21-HLA-E cells. *KLRC1*-CAR NK cells have the ability to kill multiple M21-HLA-E target cells simultaneously. The M21-HLA-E cells are represented in red, the NK cell in green, and apoptosis events in cyan. Statistical significance was calculated using a two-tailed t test (E, G, and H), and ordinary two-way ANOVA with an uncorrected Fisher's least significant difference post-test (F).

the first 4 days. As the K562 cells are eliminated from the culture beyond day 4, the NK cells may enter a less activated state even with cytokine stimulation, which could explain the gradual decrease in editing on days 5–7. Two major differences between this study and those of the Lin and Marson groups is the length and type of activation prior to nucleofection. Regardless of whether cytokines, CD2/CD335-coated beads, or K562 feeder cells are used to activate NK cells, for 2 or 4 days, this study observes comparable knockin rates across all three systems. Collectively, these findings suggest that *ex vivo* expanded NK cells are most amenable to genome engineering 2–4 days after isolation/thaw, with day 4 being an optimal time point if feeder cells are employed to ensure a pure, feeder-free NK population.

We found using an NHEJ inhibitor and fine-tuning of the pulse code within the nucleofector to be pivotal in improving our knockin outcomes. The DNA-PK inhibitor, M3814, improved HDR outcomes by 2-fold in our system. Interestingly, these results contrast with that of the Lin group where other HDR enhancers and DNA-PK inhibitors had no effect on editing efficiencies within NK cells.[21] Despite being in the same class of drugs, M3814 and KU00060648 elicit a different effect on cell viability and editing, highlighting the need for further research on the use of pharmacological enhancers for this cell type. M3814 treatment of NK cells before electroporation did not impact cell yield in our study. A previous study observed decreased cell yield for human primary T cells electroporated when using a different, stronger pulse code, EH-115, to deliver the genome editing machinery. The sensitivity of primary NK cells to M3814 may be dependent on this electroporation stress, which is worthy of future investigation.

Surprisingly, the EH-115 program resulted in the lowest percent of mCherry^{+} cells at 13.8%, which is in contrast to the results reported by the Lin group.[21] This discrepancy may be because our system uses a linear, dsDNA template for transgene insertion instead of a plasmid. The DN-100 program resulted in knockin efficiencies of 15%, which is approximately double what was published by the Lin group for the same dsDNA template strategy.[21] Nevertheless, the success of these strategies is still partially dependent on the donor from whom the cells are sourced. In our study, though all five donors had a similar frequency of NKG2A$^+$ cells (80%–95%) in the UTF population (Figure S2A), we did not observe equivalent transgene knockin outcomes across the donors, indicating that there are other inherent factors within the cell that can partially dictate editing efficiencies.

NK cells are known to be challenging to genetically engineer due to their poor recovery post-editing.[21] Their innate lineage equips them with pattern recognition receptors (PRRs) and Toll-Like receptors (TLRs) that can recognize cytosolic viral and dsDNA molecules, thus increasing dsDNA-associated toxicities.[15,16,47] We observe slower proliferation of the *KLRC1*-CAR NK and *KLRC1*-No CAR NK cells in the first 5 days following electroporation, compared with the *KLRC1*-KO and UTF NK cells. The co-delivery of a dsDNA template along with the RNP complex is likely toxic to the cells and may cause activation of cGAS-STING, AIM2, or Type I-IFN signaling, ultimately suppressing cell proliferation and inducing apoptosis. Given that our templates for PCR for dsDNA template production are transformed and grown in *E. coli,* and purified in-house, it is also possible that our templates contain trace amounts of endotoxins that may further exacerbate the dsDNA-associated toxicity. Despite these challenges, our study demonstrates that this toxicity can be overcome through careful testing of parameters with highly purified reagents.

Our work provides compelling evidence that the CRISPR platform can be utilized to generate functional CAR NK cells using a primary cell source. We observed superior cytotoxicity by the *KLRC1*-CAR NK cells compared with CAR$^-$ populations against both a melanoma and neuroblastoma cell line, indicating that the improvement in NK killing is not unique to interaction with one cell line, and points to the impact of CAR activity in driving better NK cell function. Our data indicate that NKG2A KO and CAR expression have more of an additive effect on NK potency against solid tumors than either modification by itself, with *KLRC1*-CAR NK cells displaying the greatest target lysis and IFNγ secretion compared with all other groups, in both wild-type and the HLA-E-expressing M21 system. In the HLA-E-expressing model, the difference between *KLRC1*-CAR and UTF NK cells is further widened even when cultured at a lower effector-target ratio (9% for 5:1 E:T [non-HLA-E] vs. 17% for 1:1 E:T [HLA-E]), underscoring the importance of disrupting the NKG2A-HLA-E checkpoint axis. Moreover, the upregulation of the activating NKG2C receptor, which has also been shown to bind HLA-E, in combination with greater killing per single cell, could also be a contributing factor to the improved cytotoxicity of the *KLRC1*-CAR NK cells when compared with the other groups. Further improvements in CAR NK cell cytotoxicity can be achieved by using CAR designs optimized for stronger NK cell activation, or by knocking out other known NK cell checkpoints, such as TIGIT and transforming growth factor ß signaling, through multiplex editing. Future studies could also incorporate armored CARs that can secrete critical cytokines, such as IL-15 and IL-21, as this approach has been shown to enhance CAR NK cell persistence and potency *in vivo* in several xenograft models.[36,48,49]

Clinical translation of this cell therapeutic strategy is likely viable with further research along two principal directions. First, the use of feeder cells could be eliminated through the use of feeder-free NK expansion protocols.[21,50,51] While our study utilizes feeder-based approaches to efficiently activate and expand the NK cells during manufacturing, it is important to note that feeder-free approaches would likely impact editing efficiencies and cell yield, thus requiring re-optimization of the optimal activation time, cargo delivery, pulse programs, and cytokine treatments. Second, the potency of these cells upon cryopreservation and thawing will also need to be evaluated to truly enable an off-the-shelf strategy. Future studies would need to be conducted across various animal models, small and large, to thoroughly examine the safety, efficacy, and toxicity of the engineered product, while also establishing a large-scale manufacturing protocol that aligns with cGMP guidelines. Altogether, we demonstrated the feasibility of using the CRISPR system to genetically modify primary NK cells, and of driving CAR expression and activity via the *KLRC1* locus. This CRISPR-based platform can enable virus-free manufacturing of off-the-shelf, genetically engineered NK cells to remove inhibitory immune checkpoints while simultaneously inserting CARs with the potential for enhanced potency against solid tumors.

MATERIALS AND METHODS

Cell lines

M21 human melanoma cells were a gift from Dr. Paul Sondel (University of Wisconsin-Madison), and K562-mb15-41BBL cells were a gift from Dr. Dario Campana (National University of Singapore). Both cell lines were maintained in RPMI 1640 (Gibco, Waltham, MA) supplemented with 10% fetal bovine serum (FBS) (WiCell, Madison, WI), 1% penicillin-streptomycin (Gibco, Waltham, MA), and 1% L-glutamine (Gibco, Waltham, MA). NK-92 cells (ATCC, Manassas, VA) were cultured in X-VIVO 10 (Lonza, Basel, Switzerland) supplemented with 5% FBS (WiCell, Madison, WI), 5% horse serum (Thermo Fisher, Waltham, MA), 199.82 μM myo-inositol (Sigma-Aldrich, St. Louis, MO), and 19.94 μM folic acid (Sigma-Aldrich, St. Louis, MO). The M21-HLA-E transduced cell line was maintained in the same medium with the addition of 600 μg/mL of G418 Geneticin. Cell authentication was performed using short tandem repeat analysis (Idexx BioAnalytics, Westbrook, ME) and per ATCC guidelines using cell morphology, growth curves, and *Mycoplasma* testing within 6 months with the MycoStrip Mycoplasma Detection Kit (Invitrogen, Waltham, MA). Cell lines were maintained in culture at 37°C in 5% CO_2.

Transduction of M21 cell line

The human HLA-E gene insert (Table S1), incorporating the HLA-G leader peptide, and *HindIII* and *EcoRI* restriction sites, was generated synthetically by IDT (IDT, Newark, NJ). The pcDNA3.1(+) plasmid (Thermo Fisher, Waltham, MA) and the synthesized HLA-E gene were digested separately using *HindIII* and *EcoRI* enzymes (Thermo Fisher, Waltham, MA) following the manufacturer's instructions. The digested products (backbone of pcDNA3.1 and HLA-E gene product) were ligated using the T4 DNA ligase enzyme (Thermo Fisher, Waltham, MA) according to the manufacturer's instructions. The ligated product was transformed into JM109 competent *E. coli*. (Promega, Madison, WI) using a heat shock protocol. To confirm successful transformation of clones, DNA sequencing was performed using primers specific to the pcDNA3.1 CMV (forward) and BGH (reverse) sequences.

Twenty-four hours before transfection, M21 cells (1×10^6 cells per well) were plated in a six-well plate, with cells reaching 80%–90% confluency at the time of transfection. Cell transfection was performed using the TransIT-Lenti transfection reagent (Mirus bio, Madison, WI) per the manufacturer's instructions. After 24 h of transfection, the cells were treated with 600 μg/mL of G418 Geneticin (Thermo Fisher, Waltham, MA) to select for transfected cells. Monoclonal cell populations were generated through a serial dilution technique, and flow cytometry was used to select the clone with the highest HLA-E expression.

Isolation and expansion of NK cells

Whole blood was collected from the median cubital vein of healthy donors using 10 mL lithium heparin-coated vacutainers (158 USP units, BD Biosciences, Franklin Lakes, NJ). Peripheral blood (PB) NK cells were isolated from whole blood using a density-based negative selection kit according to the manufacturer's instructions (RosetteSep Human NK Enrichment Cocktail, STEMCELL Technologies, Vancouver, Canada). Isolated PB-NK cells were expanded with irradiated K562-mb15-41BBL feeder cells (1–3 days before NK cell isolation, 100 Gy) along with 50 IU/mL IL-2 (Peprotech, Cranbury, NJ) and 2 ng/mL IL-15 (Peprotech, Cranbury, NJ) at a ratio of 1:2 in NK MACS media (Miltenyi Biotec, Gaithersburg, MD) supplemented with 5% human AB serum (Millipore Sigma, Burlington, MA) and 1% MACS supplement (Miltenyi Biotec, Gaithersburg, MD). Cells were cultured until day 4, after which they were nucleofected for experiments. After day 5, unedited and nucleofected NK cells were cultured in complete NK MACS medium (Miltenyi Biotec, Gaithersburg, MD) with 100 IU/mL IL-2 and 2 ng/mL IL-15. Cells were maintained at a density of 1e6 cells/mL and were passaged every 2 days. Feeder cells were added to support NK expansion on days 0 and 5. All human studies were conducted using a University of Wisconsin-Madison institutional review board-approved protocol (2018-0103, approved 08/31/23).

*Sp*Cas9 RNP complexing

Purified sNLS-*Sp*Cas9-sNLS Cas9 protein (10 mg/mL; Aldevron, South Fargo, ND) and sgRNA (60 pmol; IDT, Newark, NJ) with the crRNA and tracrRNA pre-complexed were aliquoted into individual PCR tubes for each nucleofection replicate. Sterile-filtered 15–50 kDa poly(L-glutamic acid) (PGA) (Sigma-Aldrich, St. Louis, MO) was added to the sgRNA at 0.8:1 volume ratio as previously described.[22] Fifty picomoles of Cas9 was added to the sgRNA + PGA mixture for a final molar ratio of 1:1.2 sgRNA:Cas9. The complexes were incubated at 37°C for 15 min prior to use.

Preparation of dsDNA templates

PCR amplicons of the CAR and mCherry transgenes were generated as described previously.[34] PCR products were pooled into 600 μL and were purified using a singular round of Solid Phase Reversible Immobilization (SPRI) cleanup using AMPure XP beads (Beckman Coulter, Brea, CA) according to the manufacturer's instructions, at a 1:1 ratio. Bead incubation and elution times were increased to 5 min and 15 min, respectively. Each pooled sample was eluted in 30 μL of nuclease-free water (IDT, Newark, NJ), which were then combined to form one sample. Three-molar pH 5.5 sodium acetate (NaOAc) (Sigma-Aldrich, St. Louis, MO) was added to the pooled sample at 0.1× of the total volume. The sample was ethanol precipitated by adding 2.5× of 100% ethanol, followed by a 30-min incubation at −80°C. The precipitate was purified using an ethanol wash (70%–80%) and eluted in 20–30 μL of nuclease-free water. DNA template concentration was measured and adjusted as previously described.[34] Primers and DNA sequences are listed in Table S1.

Nucleofection

NK cells were nucleofected on day 4, unless otherwise described for optimization experiments. Post-nucleofection recovery medium consisting of complete NK MACS medium (Miltenyi Biotec, Gaithersburg, MD) with 100 IU/mL IL-2 and 2 ng/mL IL-15 ± 0.6 μM M3814 (Selleck Chemicals, Houston, TX) was prepared, with M3814 added at 1.2X the working concentration to account for future dilution. For M3814-related optimization experiments, multiple aliquots of post-recovery medium were prepared with a corresponding concentration of M3814. Next, 150 μL of M3814 recovery medium was plated into individual wells of a 96-well round bottom plate for each replicate and set aside in the incubator. PB-NK cells were harvested and counted using the Countess II FL Automated Cell Counter (ThermoFisher, Waltham, MA) with 0.4% Trypan Blue viability stain (Thermo Fisher, Waltham, MA). 5e5 cells were aliquoted into individual 1.5-mL Eppendorf tubes for each nucleofection replicate. While the RNP complexes incubated, the cells were pelleted at $100 \times g$ for 10 min, and 3 μg (1.5 μL) of dsDNA templates were aliquoted into PCR tubes for each replicate. Following incubation, the RNP complexes were added to the dsDNA templates and were allowed to incubate for 3–5 min at room temperature. Cell pellets were resuspended in P3 buffer with 22% supplement (Lonza, Basel, Switzerland), and combined with the RNP+dsDNA mixture, for a total volume of 24 μL. Samples were added to the 24 μL reaction cuvettes (Lonza, Basel, Switzerland) and were nucleofected using the EH-100 program (Lonza Amaxa 4D Nucleofector, Basel, Switzerland), unless specified otherwise for optimization experiments. Immediately after nucleofection, 100 μL of pre-warmed recovery medium (no M3814) was added to each of the cuvette wells, and samples were rested at 37°C for 15 min. Finally, samples were transferred from the cuvette to the 96-well plate containing pre-warmed recovery medium with M3814. After 24 h, nucleofected NK cells were combined with irradiated K562-mb15-41BBL (100 Gy) at a 1:2 ratio in complete NK MACS medium with 100 IU/mL IL-2 and 2 ng/mL IL-15.

Flow cytometry and cell sorting

1e5 cells were plated per condition and stained in 96-well round bottom plates. Cells were washed with 1X PBS, centrifuged at 1,200 × *g* for 1 min, and stained with 0.1 μL/mL Zombie Aqua viability stain (Biolegend, San Diego, CA) for 20 min. Next, cells were washed twice with 2% FACS buffer (1X PBS and 2% BSA) centrifuged and stained with human TruStain FcX (BioLegend, San Diego, CA) for 20 min. Cells were washed twice more with 2% FACS buffer, pelleted, and stained with NKG2A (Clone S19004C, BioLegend, San Diego, CA) and anti-14G2a (Clone 1A7, National Cancer Institute, Biological Resources Branch, Rockville, MD) antibodies for detection of NKG2A and CAR scFv, respectively. For phenotyping experiments, NK cells were also stained with CD56 (Clone HCD56, BioLegend, San Diego, CA), CD16 (Clone 3G8, BioLegend, San Diego, CA), and NKG2C (Clone 134591, BD Biosciences, Franklin Lakes, NJ). Flow cytometry was performed on an Attune NxT flow cytometer (Thermo Fisher, Waltham, MA) and data were analyzed using FlowJo.

For sorting, NK cells were either stained for NKG2A or GD2 CAR and were sorted using a 130-μm nozzle on an FACS Aria II sorter (BD Biosciences, Franklin Lakes, NJ). Sorted cells were recovered in complete NK MACS medium (Miltenyi Biotec, Gaithersburg, MD) with 100 IU/mL IL-2 and 2 ng/mL IL-15 for 4 days prior to cytotoxicity assays. NK cells were assayed by flow cytometry or sorted 7 days after nucleofection (day 11). For flow analysis of the M21 melanoma cell line, cells were detached using 0.05% Trypsin (Thermo Fisher, Waltham, MA) for 5 min prior to filtering with a 70-μm strainer, counting and plating for staining. M21 cells were stained as described above, but using an anti-GD2 antibody (Clone 14G2a, BioLegend, San Diego, CA).

LDH release cytotoxicity assay

First, 1e4 M21 or M21-HLA-E cells were plated per well in a 96-well, flat-bottom plate in culture medium with or without 100 ng/mL of IFNγ. Twenty-four hours later, sorted NK cells were added to the well at the corresponding effector:target ratios in complete NK MACS medium supplemented with 100 IU/mL IL-2 and 2 ng/mL IL-15 for a total volume of 100 μL. For ADCC conditions, 500 ng/mL of anti-GD2 hu14.18K322A (a gift from Paul Sondel) was added to each of the wells. After 24 h of co-culture, the plate was centrifuged at 300 × *g* for 5 min to pellet the cells. Fifty microliters of supernatant was harvested and assayed for secreted lactate dehydrogenase (LDH) using the CyQuant LDH Cytotoxicity Assay kit according to the manufacturer's instructions (Thermo Fisher, Waltham, MA). The remaining supernatant was collected and frozen at −80°C for subsequent ELISA.

IFNγ ELISA

Quantification of secreted IFNγ was carried out using an IFNγ sandwich ELISA kit (Thermo Fisher, Waltham, MA) according to the manufacturer's instructions on media supernatants (undiluted) taken from the 24-h LDH co-culture assay.

Time-lapse imaging

M21-HLA-E and NK cells were stained using DiD (Thermo Fisher, Waltham, MA)and DiI (ThermoFisher, Waltham, MA), respectively. A 1:1,000 dilution of each stain was prepared in PBS and cells were stained with 1 mL each for 15 min. M21-HLA-E cells were seeded in a 96-well flat-bottom plate at a density of 30k cells/well 24-h prior to starting the time-lapse imaging. NK cells were added in conjunction with the apoptosis staining Caspase-3/7 Detection Reagent (Thermo Fisher, Waltham, MA) at a concentration of 10k NK cells/well. The apoptosis detection reagent was prepared by reconstituting in 100 μL PBS to prepare 100X stock solution. On the day of the experiment, a 10X working solution was prepared by diluting stock solution 1:10 in NK cell media. M21 cell media was aspirated and replaced with NK cell media containing CellEvent at a 1:1 ratio, and the plate was incubated for 15 min at 37°C and 5% CO_2 immediately prior to starting the time lapse. The time lapse was conducted on a Leica Thunder Microscope with stagetop incubator at 37°C 5% CO_2. Images were acquired every minute over a total of 16 h. Images were processed and analyzed using FIJI (https://imagej.net/Fiji/Downloads). Cell killing was determined by selecting representative NK cells at the beginning of the time lapse and manually tracking their activity.

IncuCyte assays

A total of 10,000 GFP-labeled CHLA-20 cells were plated per well in a 96-well, flat-bottom plate in culture medium. Twenty-four hours later, thawed and sorted NK cell products were added to the well at 1:1 and 5:1 E:T ratios in complete NK MACS medium supplemented with 100 IU/mL IL-2 and 2 ng/mL IL-15 for a total volume of 100 μL. Cell killing was measured via the IncuCyte S3 Live-Cell Analysis system, with images taken every 4 h for 24 h. Green object count was used to quantify the number of cancer cells per well. Cytotoxicity for a given sample was calculated as the difference between the green object count at a given time minus the count at t = 0 divided by the count at t = 0. Image analysis was performed using the IncuCyte Analysis Software.

In-out PCR

Genomic DNA was extracted from 1e6 cells as described previously.[34] One primer was designed to bind upstream of the left homology arm or downstream of the right homology arm, depending on the orientation of transgene integration. The second primer was designed to bind within the scFv region of the CAR, or within the protein coding region of the mCherry transgene. PCRs were carried out using the Q5 Hot Start 2× Master Mix (NEB, Ipswich, MA) according to the manufacturer's protocol for 30 cycles. Primer sequences are listed in Table S1.

On-target editing analysis via NGS

Genomic DNA was extracted and indel formation was analyzed via NGS (Illumina, San Diego, CA) as previously described.[34] Briefly, edited amplicons were amplified and purified using genomic PCR and SPRI cleanup, respectively. A second PCR was performed with indexing primers to attach unique dual identifiers (Illumina, San Diego, CA) to the amplicons. After a second SPRI cleanup, samples were combined and run on the Illumina MiniSeq per the manufacturer's instructions and analyzed using the CRISPR RGEN and CRISPResso2[52] software. Primer sequences are listed in Table S1.

Off-target nomination via genome-wide CHANGE-seq

The Gentra Puregene kit (Qiagen, Germantown, MD) was used to extract genomic DNA from unedited primary NK cells per the manufacturer's instructions. Genomic analysis was performed by CHANGE-seq as previously described.[53] In short, genomic DNA underwent tagmentation with a custom Tn5-transposome, and was gap repaired with HiFi HotStart Uracil+ Ready Mix (Kapa Biosystems, Wilmington, MA). USER enzyme (NEB, Ipswich, MA) and T4 polynucleotide were then used to treat the gap-repaired DNA. DNA circularization was performed using T4 DNA ligase (NEB, Ipswich, MA), followed by treatment with a mixture of exonucleases to remove residual linear DNA. *In vitro* cleavage reactions were performed on the circularized DNA using *Sp*Cas9 protein (NEB, Ipswich, MA) and *KLRC1* sgRNA. Cleaved products were incubated with proteinase K (NEB, Ipswich, MA), A-tailed, ligated with a hairpin adapter (NEB, Ipswich, MA), and treated with USER enzyme (NEB, Ipswich, MA). Products were amplified by PCR with the Kapa HiFi polymerase (Kapa Biosystems, Wilmington, MA), and libraries were quantified by qPCR (Kapa Biosystems, Wilmington, MA). Libraries were sequenced with 150 bp paired-end reads on the Illumina NextSeq2000 instrument. CHANGE-seq data analyses were performed as described previously.[53]

Off-target analysis at nominated sites via NGS

Indel frequency at off-target sites was assayed by the rhAmpSeq system (IDT, Newark, NJ). Genomic DNA was extracted from *KLRC1*-CAR and UTF NK cells. Pools of forward and reverse primers were designed using IDT's rhAmpSeq design tool to amplify the on-target and top 12 off-target sites detected by CHANGE-seq. PCRs and library preparation were performed according to IDT's instructions. Pooled libraries were diluted and sequenced on the Illumina MiniSeq instrument with 150 bp paired-end reads. Data were analyzed using IDT's rhAmpSeq analysis tool.

Integration site analysis via long-read WGS

Genomic DNA from *KLRC1*-CAR NK cells was extracted using the Gentra Puregene kit (Qiagen, Germantown, MD) per the manufacturer's instructions, and quantified using Qubit 3.0 Fluorometer (Thermo Fisher, Waltham, MA). Sequencing libraries were prepared using the ONT Ligation Sequencing Kit V14 (SQK-LSK114, Oxford, UK) according to the manufacturer's instructions with minimal modifications (namely, the gDNA was sheared using a Covaris g-tube prior to library preparation (Covaris, Woburn, MA), doubled FFPE and end-repair time, and doubled the amount of beads during clean up), and sequenced on the ONT PromethION instrument (Flowcell #: FLO-PRO114M) (Oxford, UK).

WGS reads were aligned to the scFv region of the CAR transgene; reads with alignment to the transgene were subsequently mapped to the human genome (GRCh38). Mapping was performed with minimap2 utilizing the first half of the detectIS pipeline.[54,55] Reads were filtered at a mapq threshold of 30. Integration sites were identified from reads aligning to both the transgene and the human genome filtered with an overlap window threshold (distance between the transgene aligning and the human genome aligning segments/total length of read) of 0.15.

Statistical analysis

All analyses were performed using GraphPad Prism (V10.1.0). For experiments with two groups, a two-tailed t test was used. For optimization experiments with more than two groups, a one-way ANOVA was used, with the Prism recommended post-test. For editing and cytotoxicity experiments with multiple donors, a one-way or two-way ANOVA was used, followed by the Prism recommended post-test. Post-tests are listed in the figure legend for each experiment. Error bars represent mean (SD). ns, $p \geq 0.05$; $^{*}p \leq 0.05$; $^{**}p \leq 0.01$; $^{***}p \leq 0.001$; $^{****}p \leq 0.0001$.

DATA AVAILABILITY

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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AUTHOR CONTRIBUTIONS

K.Shankar and B.E.R. designed the sgRNA sequences, and K.Shankar designed the mCherry and CAR sequences. K.Shankar and B.E.R. designed, performed, and analyzed KO experiments and NGS studies in NK-92 cells. K.Shankar collected whole blood, isolated NK cells, and established the cell expansion and gene editing protocol. K.Shankar designed, performed, and analyzed all gene editing optimization studies and co-culture experiments. I.Z.-H. assisted with DNA template production, cell culture, extraction of genomic DNA, and nucleofections. DMT and DM assisted with isolations, nucleofections, and cell culture for functional experiments. S.Z. performed serial killing assays and analyzed the corresponding data. L.S. and M.H.F. generated the M21-HLA-E line and helped perform characterization experiments and functional experiments for manuscript revisions. S.Z.D. performed long-read whole-genome sequencing. K.G. completed the analysis for the whole-genome sequencing study. V.K. performed and analyzed the CHANGE-seq assay and K.Shankar performed PCRs, carried out NGS experiments, and ran analysis for subsequent off-target experiments. K.Shankar wrote the manuscript with input from all authors. K.Saha, C.M.C., S.Q.T., M.M., and J.M.A. supervised the research. All authors read and approved the final manuscript.

DECLARATION OF INTERESTS

K.Saha receives honoraria for advisory board membership for Andson Biotech, Bharat Biotech, and Notch Therapeutics. C.M.C. receives honoraria for advisory board membership for Bayer, Elephas, Nektar Therapeutics, Novartis, and WiCell Research Institute. S.Q.T. receives honoraria and equity from scientific advisory board membership for Prime Medicine and Ensoma. S.Q.T. is one of the co-inventors in a patent describing

CHANGE-seq. M.M. consults for the Translational Genomics Research Institute (TGen), a not-for-profit research institution. K.Shankar and K.Saha are inventors on a patent application to be filed with the Wisconsin Alumni Research Foundation (WARF) for the technology in this study.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.ymthe.2025.01.024.

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